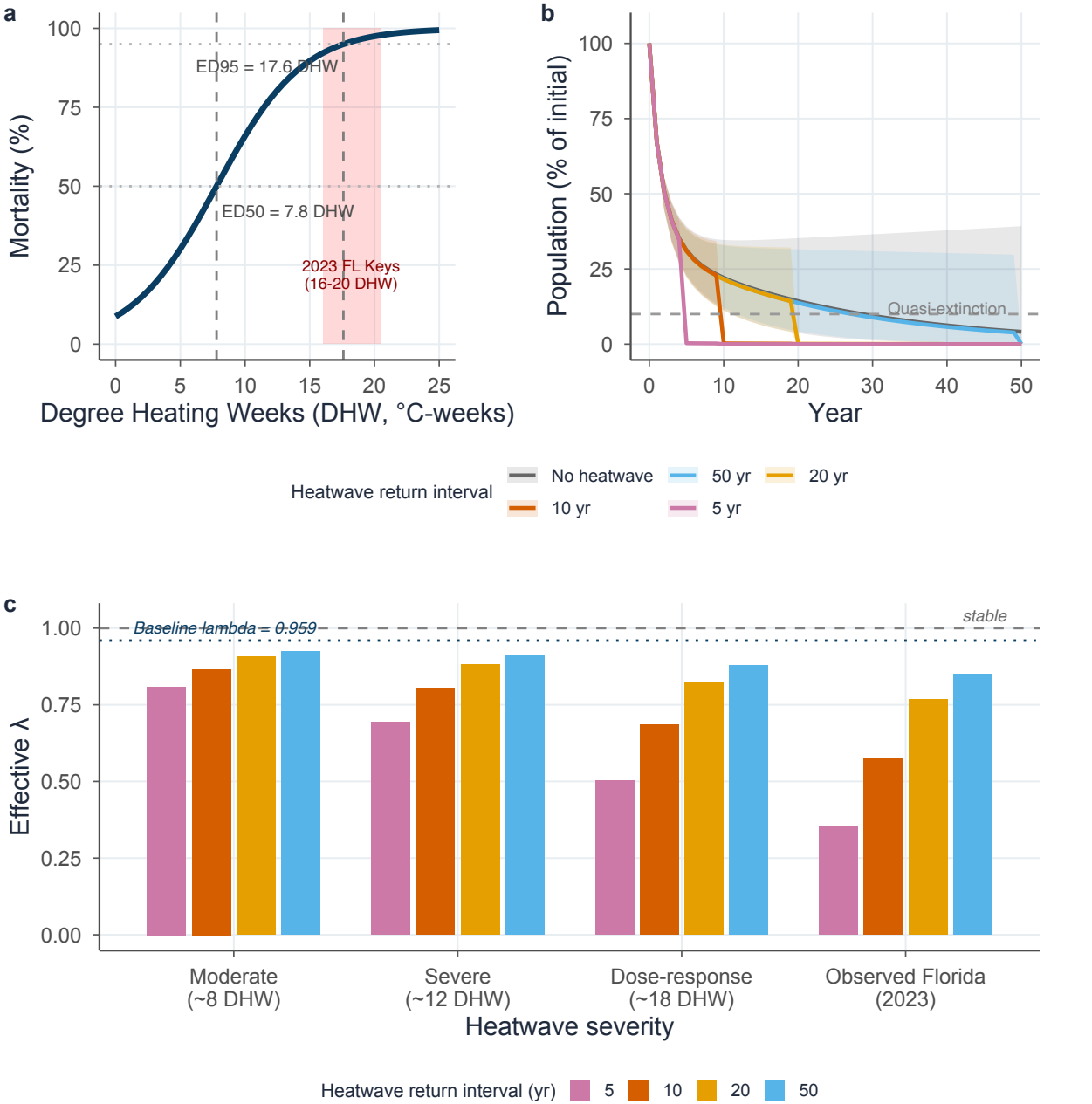

a
Mortality (%)
100
75
50
25
0
ED95 = 17.6 DHW
ED50 = 7.8 DHW
2023 FL Keys
(16-20 DHW)
0
5
10
15
20
25
Degree Heating Weeks (DHW, °C-weeks)
b
Population (% of initial)
100
75
50
25
0
Quasi-extinction
0
10
20
30
40
50
Year
Heatwave return interval
No heatwave
50 yr
20 yr
10 yr
5 yr
c
Baseline lambda = 0.959
stable
Effective λ
1.00
0.75
0.50
0.25
0.00
Moderate
(~8 DHW)
Severe
(~12 DHW)
Dose-response
(~18 DHW)
Observed Florida
(2023)
Heatwave severity
Heatwave return interval (yr)
5
10
20
50